

Challenges in Implementing Post-Quantum Cryptography Across Modern Infrastructure

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Abstract

Quantum computing presents a serious challenge to modern cryptographic systems, especially those built on problems like integer factorization and discrete logarithms. In response, the National Institute of Standards and Technology (NIST) introduced post-quantum cryptographic (PQC) standards in 2024, including ML-KEM, ML-DSA, and SLH-DSA, which are designed to remain secure against both classical and quantum attacks. While these algorithms provide a strong theoretical solution, their real-world implementation is far more complex. This paper reviews recent research, including benchmarking studies, protocol integration experiments, and institutional guidance, to better understand the gap between PQC's theoretical readiness and practical deployment. The analysis focuses on four areas: protocol integration, certificate infrastructure, constrained devices, and organizational preparedness. Across these domains, the findings show that PQC works in controlled environments but introduces performance overhead, compatibility issues, and hardware limitations that prevent widespread deployment. In addition, the lack of standardized certificate models and limited enterprise readiness further slows adoption. Overall, the challenge is no longer about developing secure algorithms, but about integrating them into existing systems. Addressing this gap is critical to ensuring a smooth and timely transition to quantum-resistant security.

Keywords: post-quantum cryptography (PQC), quantum computing, cryptographic implementation, network protocols, constrained devices, certificate infrastructure

Introduction

Modern cryptography is the foundation for security of nearly every corner of the internet, from digital transactions and authentication to VPNs and communications, all relying on asymmetric and symmetric standards. Symmetric cryptography derives its security through sheer

key size and the complexity of ciphers, whereas asymmetric cryptography, also known as public key cryptography, derives its security from two mathematically related keys. RSA and elliptic curve cryptography (ECC) are the two most widely embedded algorithms in today's infrastructure. These depend on the computational difficulty of integer factorization and discrete logarithm problems. However, the increasing development of quantum computers now presents a significant threat to these cryptographic methods. With quantum algorithms such as Grover's and Shor's, longstanding cryptographic assumptions are challenged: Grover's algorithm reduces the effective security of symmetric cryptography, while Shor's algorithm solves the mathematical foundations of public-key cryptography in polynomial time, rendering current Public-Key Infrastructure obsolete against a sufficiently capable quantum system.

NIST responded with a standardization effort launched in 2016, culminating in the August 2024 publication of ML-KEM (FIPS 203), ML-DSA (FIPS 204), and SLH-DSA (FIPS 205). These are the first internationally recognized quantum-resistant standards designed to withstand both classical and quantum attacks in the current infrastructure. The NSA has noted that new cryptography can take twenty years or more to fully deploy across national security systems, and NIST's own migration guidance acknowledges that most organizations lack complete visibility into where cryptography is even used across their infrastructure.

The publication of FIPS 203, 204, and 205 resolved the question of which algorithms are mathematically sound against quantum attacks. However, there is a second, equally serious problem: how these algorithms are deployed across the heterogeneous infrastructure that runs the internet. That infrastructure includes network protocols unprepared for larger key sizes, a global X.509 certificate ecosystem that requires fundamental restructuring, and billions of constrained devices that cannot absorb the computational overhead that PQC algorithms impose. Although

NIST's 2024 post-quantum standards resolve the algorithmic question, the challenge is now deployment: the existing infrastructure is fundamentally unprepared to handle quantum-resistant cryptography. This paper analyzes peer-reviewed benchmarking studies, protocol integration experiments, and institutional guidance to examine the implementation challenge across four domains: protocol integration, certificate infrastructure, constrained devices, and the institutional response that frames the urgency of the transition.

Literature Review

Institutional Awareness of the Quantum Threat

The urgency of post-quantum migration is well-documented at the institutional level, though the gap between awareness and action remains striking. The NSA (2021) frames the timeline problem clearly: because new cryptography can take 20 years or more to fully deploy across national security systems, the absence of a confirmed CRQC is not a reason to delay. Le et al. (2025) quantify exactly how wide that gap is — fewer than 5% of enterprises have a formal quantum transition plan despite 62% of technology professionals expressing concern about quantum threats. NIST SP 1800-38 reinforces this concern from a practical standpoint, acknowledging that many organizations lack complete visibility into how cryptography is used across their systems and that discovering the full scope of cryptographic dependencies will itself take considerable time. Together, these sources converge on a consistent finding: institutional awareness has not translated into institutional readiness, and the migration timeline demands that it does.

Protocol Integration

Standardizing PQC algorithms is a must for implementing quantum-safe cryptography, but it's not easy, as it raises questions about what happens when these algorithms are deployed in

real network infrastructure. Across peer-reviewed research, protocols consistently show that PQC works in controlled environments but lack the foundation for steady production.

Aydeger et al. (2025), published in the *Infocommunications Journal*, integrate ML-KEM-512 into the DNS-over-TLS protocol to evaluate whether PQC can be deployed at this foundational layer. DNS enables devices to resolve domain names into their IP addresses billions of times per second across global infrastructure, and because DNS-over-TLS encrypts those queries using a TLS session built on asymmetric key exchange, every DNS query becomes a point of quantum vulnerability. Aydeger et al. find that the TLS handshake phase is the dominant performance bottleneck under PQC, with overhead increasing as query loads increase. The authors acknowledge that results at the small scale are encouraging but are wary of their applicability to real-world production. They find PQC introduces latency and computational demands that current implementations are not equipped to absorb. This finding matters because DNS sits below everything else, meaning that overhead at this layer compounds upward as data moves.

Moving to the session layer, Zhang et al. (2025), published in the peer-reviewed journal *Entropy*, provide a comprehensive PQC testing framework and validate it through testing of a hybrid X25519-Kyber768 under simulated conditions. Their results support Aydeger's handshake findings at a different protocol level. The Kyber768 key-generation process introduced end-to-end latency overhead relative to classical X25519 baselines. While the TLS 1.3 handshake was compatible, Zhang et al. frame their findings as a starting point rather than a resolution, as the study was conducted in a controlled environment.

This distinction becomes harder to ignore when PQC is applied to infrastructure management protocols rather than web traffic. Vorobets et al. (2025), presented at the Classic,

Quantum, and Post-Quantum Cryptography conference, examine post-quantum SSH via the OQS-OpenSSH experimental fork, which integrates the liboqs library to support hybrid key-exchange configurations. SSH is foundational to secure infrastructure management, and the authors demonstrate a working deployment on AWS EC2 instances using a hybrid classical and post-quantum configuration. Vorobets et al. state that these implementations are not ready for production, citing the absence of security auditing, incomplete alignment with standards, and unresolved interoperability challenges with existing infrastructure.

Where Vorobets et al. document the implementation gap of experimental feasibility and production readiness at the protocol level, Lawo et al. (2024), published in *Future Internet*, push further into the deployment stack, demonstrating what a quantum-safe IPsec tunnel looks like in practice, finding that even at this level of implementation, the path forward depends on a hardware infrastructure that does not exist. IPsec secures IP communications by encrypting and authenticating each data packet and is widely used in enterprise VPNs and data center communications. Lawo et al. benchmark performance across two scenarios: a high-throughput data center environment with 100 Gbit/s fiber-connected servers. Their findings indicate that PQC algorithms carry substantially larger key and signature sizes than their classical counterparts. The authors identify two unresolved challenges: the full integration of PQC into production software stacks and the development of dedicated hardware accelerators to offset PQC's computational demands.

Across all four studies, the same critical issue emerges: PQC can work in experimental environments but lacks production-readiness for widespread implementation.

Certificate Infrastructure Research

While protocol-level integration demonstrates that PQC can function in controlled environments, these implementations depend on a certificate infrastructure that has yet to evolve at the same pace. Cho et al. (2025), published by researchers at Hansung University and Nanyang Technological University, analyze 192 live TLS sessions across 20 major domestic and global enterprises using an automated packet inspection system that extracts cipher suite, key exchange, and certificate data without requiring server-side decryption. The global enterprises showed TLS 1.3 adoption of 90% and PQC hybrid key exchange adoption of 40%, while domestic Korean enterprises showed 40% and 0%, respectively. Across all 192 observed sessions, certificate infrastructure remained entirely classical. This means that organizations with well-documented PQC research have not made the strides for certificate model replacements in live environments.

Yusof et al. (2026), published in the *Malaysian Journal of Science, Health and Technology*, provide a systematic review of 27 studies published between 2017 and 2025, identify six distinct post-quantum certificate models, and find that no consensus has emerged around any of them. The Hybrid model, which combines classical and post-quantum components within a single X.509 structure, is id Cho et al. and Yusof et al. point to the same underlying problem from different angles: the certificate infrastructure question is simultaneously a standardization problem and a deployment problem, and neither has been resolved.

Constrained Device Research

Constrained device research introduces a hardware dimension that software-level solutions cannot fully address. Fitzgibbon and Ottaviani (2024) establish the baseline problem — PQC algorithms on Raspberry Pi 4 hardware produce measurable TLS 1.3 handshake latency degradation even under the optimized liboqs library — suggesting that algorithm selection alone

does not account for the computational limitations of resource-constrained environments. Algar-Fernandez et al. (2026), published in *Computers*, extend that finding in a direction that makes the constrained device problem harder to solve. Their controlled benchmarking study evaluates eight PQC primitives across two processor architectures, AMD Ryzen 7 4000 and Intel Core i5-1035G1. The AMD platform consistently outperforms Intel, showing that the same algorithm incurs different overheads depending on the hardware architecture. This shows a complex implementation problem, as standardized PQC algorithms have different effects across competing architectures. Khan et al. (2024) extend this finding to consumer electronics, documenting that optimized PQC introduces manageable overhead on server hardware while running up to 12 times slower on consumer devices. Mahdi and Abdullah (2025) and KEBANDE (2025) both point toward the same conclusion finding that while lattice-based and hash-based approaches are the most viable candidates, incorporating them into existing constrained and industrial infrastructure remains an open research problem. Across these five studies, the constrained device literature is notably consistent: the computational requirements of current PQC algorithms exceed what much of the deployed device ecosystem can absorb, and hardware acceleration is identified as a necessary condition rather than an optional optimization.

Methodology

This paper employs a systematic literature review to examine the implementation gap between post-quantum cryptography and practical deployment across real-world infrastructure. Rather than evaluating the mathematical properties of PQC algorithms, this review focuses on the challenges organizations face when attempting to deploy those algorithms in existing systems. Those challenges are organized into four domains: protocol integration, certificate infrastructure, constrained device performance, and institutional readiness.

Sources were gathered through the Gardner-Webb University Online Library and Google Scholar. Search terms included combinations of "post-quantum cryptography implementation," "PQC protocol integration," "quantum-safe TLS," "post-quantum certificates X.509," "PQC constrained devices," "PQC IoT benchmarking," "crypto-agility," and "quantum cryptography migration." Institutional documents were retrieved directly from the NIST and NSA websites and used to establish that the mathematical foundation of the selected algorithms is sound. A source had to meet three criteria: it had to address real-world deployment of PQC rather than purely theoretical analysis, it had to be published between 2024 and 2026 to reflect the rapidly evolving state of the field following NIST's August 2024 standardization, and it had to be a peer-reviewed journal article. The sources were examined for its findings on feasibility, performance overhead, backwards compatibility and the barrier preventing deployment. Taken all together, these articles fundamentally work to prove a commonality of implementation barriers among different domains.

Results

The systematic review of peer-reviewed benchmarking studies, protocol integration experiments, and live traffic measurements finds that PQC algorithms are experimentally functional across every examined protocol layer and device context but there is a gap before widespread implementation. The barrier behind real-world implementation is documented independently across DNS, TLS, SSH, IPsec, certificate infrastructure, and constrained device environments, with each domain presenting a distinct barrier to widespread transition.

Across the four examined protocol studies, experimental success is accompanied by performance overhead, infrastructure incompatibility, or an author classification of its non-production status. Aydeger et al. report handshake latency increases of approximately 250% at

400 DNS queries and 500% at 800 queries with ML-KEM-512 enabled, with handshake-specific throughput dropping 80 to 82%, while network and PQC operation latency remain stable. Their research isolates the handshake as the single failure point under load. Zhang et al. examine this at the TLS session layer, where the client Key_exchange value expands from 32 bytes under classical X25519 to 1,216 bytes under hybrid X25519Kyber768, and end-to-end latency increases significantly despite confirmed handshake conformance and 12-hour stability under sustained load. At the infrastructure management layer, Vorobets et al. confirm hybrid key exchange operational on AWS EC2 but explicitly classify the implementation as not production-ready, citing absent security auditing, incomplete standards alignment, and unresolved interoperability with existing SSH deployments, while identifying static RSA keys with no rotation policy as a structural exposure across the SSH ecosystem. Lawo et al. demonstrate 100 Gbit/s throughput in a fiber data center IPsec scenario and 0.486 Gbit/s in a wireless mobile scenario, but point to two explicitly unresolved production barriers: full PQC integration into existing software stacks and the absence of dedicated PQC hardware accelerators, with Falcon key generation requiring approximately two orders of magnitude more CPU clock cycles than Dilithium and Dilithium signatures reaching 4,595 bytes at the highest security level compared to 256 bytes for RSA-2048.

The certificate infrastructure research shows that the disconnect between protocol-layer and certificate-layer PQC adoption is not marginal. Cho et al.'s live measurement of 192 TLS sessions across 20 major enterprises finds that 40% of global enterprises have deployed PQC hybrid key exchange at the protocol layer, while 0% of enterprises in either group have deployed post-quantum certificates. This finding is drawn from organizations including Google, YouTube, PayPal, and Instagram. Yusof et al.'s systematic review of 27 studies shows that across six

identified post-quantum certificate models, no model has achieved production consensus; ML-DSA signatures exceed 2 kilobytes compared to under 300 bytes for classical counterparts, TLS handshake times in IoT deployments can double with PQC-only certificates, and most commercial certificate authority platforms lack native PQC support.

Across constrained device environments, five independent studies find that none of the standardized PQC algorithms satisfy the full set of constraints imposed by physical hardware. Fitzgibbon and Ottaviani document performance degradation across all tested algorithms on Raspberry Pi 4 devices. Algar-Fernandez et al. find that the same algorithms impose different overhead depending on processor architecture, with AMD Ryzen 7 consistently outperforming Intel Core i5 across all tested primitives, and none performing reliably across key size, signature size, and computational cost. Khan et al. report that optimized PQC algorithms run up to 12 times slower on consumer devices than on server hardware and characterize the problem as a hardware infrastructure limitation without a current solution. Mahdi and Abdullah find that no existing algorithm satisfies the combined IoT constraints of low computation, small key size, low memory, and low energy. Kebande identifies long device replacement cycles, strict uptime requirements, and legacy system incompatibility as barriers specific to industrial IoT, while classifying integration of PQC into existing industrial infrastructure as an open research problem. Across consumer, IoT, and industrial IoT contexts, the finding is that the hardware infrastructure required for practical PQC deployment is not here yet.

Discussion

The research provided throughout this paper demonstrates the NIST's 2024 standardization addressed the algorithm question, it did not resolve the looming deployment

question. The two are documented as being very distinct issues, but the research reviewed shows that implementation may fundamentally be harder to tackle.

At the protocol level, the handshake overhead documented by Aydeger et al. and Zhang et al. show that this is a direct consequence of larger key sizes, which are themselves a mathematical property of the algorithms. This means the performance costs observed in experimental settings are not expected to disappear as implementations mature but are the bare minimum that entire infrastructures will need to be redesigned to support. The results from Vorobets et al. and Lawo et al. show that even where PQC is technically operational, the surrounding infrastructure has not been built yet. Protocol integration is waiting on an ecosystem that can support it.

The certificate findings from Cho et al. demonstrate the true nature of actual deployment because the organizations in their sample are not one-off companies but Google, PayPal, and Meta. These represent some of the most sophisticated deployments on the internet and even they have not moved the needle. This shows that the transition is delayed by the lack of a standardized path forward which Yusof et al. confirm. Until the certificate layer catches up, organizations that have already upgraded their key exchange mechanisms are running a hybrid deployment. The implication of this is that the industry cannot treat this as a single migration event. It requires coordinated, layered transitions across key exchange, certificates, and protocol stacks.

The problem documented by Khan et al. and Mahdi and Abdullah is current PQC algorithms were not designed with IoT-class hardware in mind, and no existing algorithm satisfies all IoT constraints at once. This has direct consequences for sectors such as industrial control systems, medical devices, and critical infrastructure, where devices have long replacement cycles and cannot be upgraded on the timeline that quantum migration requires.

Kebande describes IIoT integration as an open research problem which is not an overstatement as it shows the genuine gap between where the algorithms are and where the hardware is.

There are two different areas for future research. One being large-scale deployments are needed across all four domains. The studies reviewed here are controlled experiments or small-scale pilots, and it is not yet known how PQC performs under the actual traffic the internet infrastructure handles. The other being the constrained device gap deserves dedicated research into PQC algorithm variants designed specifically for resource-limited environments.

What the current body of research establishes clearly is that the deployment gap is real, documented across independent domains, and will not close without effort across protocol engineering, certificate infrastructure, hardware development, and institutional planning.

Conclusion

The publication of FIPS 203, 204, and 205 in August 2024 resolved the question the cryptographic community had been working toward for nearly a decade: which algorithms can withstand a quantum attack. The harder question is how those algorithms get deployed across the modern internet. The evidence reviewed in this paper documents that gap consistently across four independent domains. Protocols introduce handshake overhead that current infrastructure was not designed to absorb. Certificate ecosystems have not moved in step with key exchange advancements, leaving even the most technically sophisticated organizations running only partially quantum-resistant deployments. Constrained devices face hardware ceilings that no existing standardized algorithm can clear within IoT resource budgets. And institutions, despite documented awareness of the threat, have largely not converted that awareness into formal transition plans.

The NSA has noted that cryptographic transitions across national security systems have historically taken 20 years or more, and NIST's own migration guidance acknowledges that most organizations do not yet have full visibility into where cryptography runs across their own infrastructure. Those two facts together show the timeline is already tight, and most organizations have not started. What this paper argues is that mathematical standardization was a necessary condition for the post-quantum transition but does not completely solve the threat. The deployment gap is real, and it will not close through algorithm development alone. Closing it will require coordinated progress across protocol engineering, certificate infrastructure, constrained device hardware, and institutional planning.

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